

Operations Research

Transportation Problem — LP Formulation & Solution

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Problem Statement

Question No. 1

- (a) Define a transportation problem in operations research. Look at the network diagram given below and develop a linear programming problem.
- (b) Discuss balanced and unbalanced transportation problems with examples.

Transportation Cost Matrix (from Network Diagram)

Source \ Dest.	D_1	D_2	D_3	Supply a_i
S_1	$c_{11} = 3$	$c_{12} = 4$	$c_{13} = 2$	5
S_2	$c_{21} = 2$	$c_{22} = 2$	$c_{23} = 3$	7
S_3	$c_{31} = 3$	$c_{32} = 1$	$c_{33} = 3$	3
Demand b_j	7	3	5	15

c_{ij} = per-unit shipping cost from source S_i to destination D_j . Each cell entry is read from the arrow labels in the network diagram.

Part (a): Definition and LP Formulation

Definition of Transportation Problem

A **transportation problem** is a special class of linear programming problem concerned with distributing a homogeneous commodity from m *sources* (origins) to n *destinations* at **minimum total cost**.

Each source S_i possesses a fixed supply a_i and each destination D_j requires a fixed demand b_j . The unit shipping cost from S_i to D_j is c_{ij} . The decision variables $x_{ij} \geq 0$ represent the quantity shipped on each route. A feasible solution must satisfy every supply and demand constraint exactly (for a balanced problem) or within bounds (for unbalanced cases).

Transportation problems arise in logistics, supply chain management, facility location, and production planning.

Decision Variables

Let x_{ij} = number of units transported from source S_i to destination D_j ($i = 1, 2, 3; j = 1, 2, 3$).

LP Formulation

Objective Function — Minimise Total Transportation Cost

$$\begin{aligned} \text{Minimise } Z = & 3x_{11} + 4x_{12} + 2x_{13} \\ & + 2x_{21} + 2x_{22} + 3x_{23} \\ & + 3x_{31} + x_{32} + 3x_{33} \end{aligned}$$

Supply and Demand Constraints

Supply constraints (total shipped from each source = its supply):

$$x_{11} + x_{12} + x_{13} = 5 \quad (S_1)$$

$$x_{21} + x_{22} + x_{23} = 7 \quad (S_2)$$

$$x_{31} + x_{32} + x_{33} = 3 \quad (S_3)$$

Demand constraints (total received at each destination = its demand):

$$x_{11} + x_{21} + x_{31} = 7 \quad (D_1)$$

$$x_{12} + x_{22} + x_{32} = 3 \quad (D_2)$$

$$x_{13} + x_{23} + x_{33} = 5 \quad (D_3)$$

Non-negativity: $x_{ij} \geq 0$ for all i, j .

Balance Check: Total Supply = $5 + 7 + 3 = 15$, Total Demand = $7 + 3 + 5 = 15$.
Supply = Demand $\Rightarrow$ **Balanced Transportation Problem.** No dummy variable needed.

Solution: North-West Corner Method (Initial BFS)

NWC Step-by-Step Allocations

Start at cell $(1, 1)$ and move right/down, always allocating $\min(\text{remaining supply, remaining demand})$.

	D_1 [7]	D_2 [3]	D_3 [5]	Supply
S_1 [5]	5 (3)	4	2	5
S_2 [7]	2 (2)	3 (2)	2 (3)	7
S_3 [3]	3	1	3 (3)	3

Bold = allocation, parentheses = unit cost.

Route	Units	Cost	Total
$S_1 \rightarrow D_1$	5	3	15
$S_2 \rightarrow D_1$	2	2	4
$S_2 \rightarrow D_2$	3	2	6
$S_2 \rightarrow D_3$	2	3	6
$S_3 \rightarrow D_3$	3	3	9
Initial Cost			40

Number of basic variables = $5 = m + n - 1 = 3 + 3 - 1$. Non-degenerate. ✓

Optimality Test: MODI Method — Iteration 1

Set $u_1 = 0$. Compute duals u_i, v_j via $c_{ij} = u_i + v_j$ for each basic cell:

$$v_1 = 3, \quad u_2 = -1, \quad v_2 = 3, \quad v_3 = 4, \quad u_3 = -1$$

Reduced cost $\Delta_{ij} = c_{ij} - u_i - v_j$ for non-basic cells:

Cell	c_{ij}	$u_i + v_j$	Δ_{ij}	Status
(1, 2)	4	$0 + 3 = 3$	+1	
(1, 3)	2	$0 + 4 = 4$	-2	Enter (most negative)
(3, 1)	3	$-1 + 3 = 2$	+1	
(3, 2)	1	$-1 + 3 = 2$	-1	

Enter (1, 3). Loop: $(1, 3)^+ \rightarrow (2, 3)^- \rightarrow (2, 1)^+ \rightarrow (1, 1)^-$

$$\theta = \min\{x_{23} = 2, x_{11} = 5\} = 2$$

Updated: $x_{13} = 2, x_{23} = 0$ (leaves), $x_{21} = 4, x_{11} = 3$. Cost = $40 - 2 \times 2 = 36$

MODI Method — Iteration 2

New BFS: $x_{11} = 3, x_{21} = 4, x_{22} = 3, x_{13} = 2, x_{33} = 3$

Duals ($u_1 = 0$): $v_1 = 3, v_3 = 2, u_2 = -1, v_2 = 3, u_3 = 1$

Cell	c_{ij}	$u_i + v_j$	Δ_{ij}	Status
(1, 2)	4	$0 + 3 = 3$	+1	
(2, 3)	3	$-1 + 2 = 1$	+2	
(3, 1)	3	$1 + 3 = 4$	-1	
(3, 2)	1	$1 + 3 = 4$	-3	Enter (most negative)

Enter (3, 2). Loop: $(3, 2)^+ \rightarrow (2, 2)^- \rightarrow (2, 1)^+ \rightarrow (1, 1)^- \rightarrow (1, 3)^+ \rightarrow (3, 3)^-$

$\theta = \min\{x_{22} = 3, x_{11} = 3, x_{33} = 3\} = \mathbf{3}$

Three cells tie; remove (2, 2). Cells (1, 1) and (3, 3) become 0 (degenerate basic variables).

Updated allocations: $x_{32} = 3, x_{21} = 7, x_{13} = 5; x_{11} = 0, x_{33} = 0$ (degenerate in basis).

Cost = $36 - 3 \times 3 = \mathbf{27}$

MODI Method — Iteration 3 (Degenerate Pivot)

BFS: $\{x_{11} = 0, x_{21} = 7, x_{13} = 5, x_{32} = 3, x_{33} = 0\}$

Duals ($u_1 = 0$): $v_1 = 3, v_3 = 2, u_2 = -1, u_3 = 1, v_2 = 0$

Cell	c_{ij}	$u_i + v_j$	Δ_{ij}	Status
(1, 2)	4	$0 + 0 = 0$	+4	
(2, 2)	2	$-1 + 0 = -1$	+3	
(2, 3)	3	$-1 + 2 = 1$	+2	
(3, 1)	3	$1 + 3 = 4$	-1	Enter

Enter (3, 1). Loop: $(3, 1)^+ \rightarrow (1, 1)^- \rightarrow (1, 3)^+ \rightarrow (3, 3)^-$

$\theta = \min\{x_{11} = 0, x_{33} = 0\} = 0$ (pure degenerate pivot; no cost change)

Remove (1, 1); replace with $(3, 1) = \varepsilon$. Cost remains **27**.

New BFS: $\{x_{31} = \varepsilon, x_{21} = 7, x_{13} = 5, x_{32} = 3, x_{33} = 0\}$

Recomputed duals: $u_1 = 0, u_2 = 0, u_3 = 1; v_1 = 2, v_2 = 0, v_3 = 2$

Cell	c_{ij}	$u_i + v_j$	Δ_{ij}	Status
(1, 1)	3	$0 + 2 = 2$	+1	
(1, 2)	4	$0 + 0 = 0$	+4	
(2, 2)	2	$0 + 0 = 0$	+2	
(2, 3)	3	$0 + 2 = 2$	+1	

All $\Delta_{ij} \geq 0 \Rightarrow$ **Optimal solution reached.**

Optimal Solution**Optimal Allocation:**

Route	Units	Unit Cost	Total
$S_1 \rightarrow D_3$	5	2	10
$S_2 \rightarrow D_1$	7	2	14
$S_3 \rightarrow D_2$	3	1	3
Minimum Transportation Cost			27

Verification: $S_1: 5 = 5\checkmark$ $S_2: 7 = 7\checkmark$ $S_3: 3 = 3\checkmark$ $D_1: 7 = 7\checkmark$ $D_2: 3 = 3\checkmark$ $D_3: 5 = 5\checkmark$

Part (b): Balanced and Unbalanced Transportation Problems**1. Balanced Transportation Problem**

A transportation problem is **balanced** when total supply equals total demand:

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$$

Balanced problems are solved directly using standard methods (NWCM, VAM, LCM + MODI).

Example (the problem above):

	D_1	D_2	D_3	Supply
S_1	3	4	2	5
S_2	2	2	3	7
S_3	3	1	3	3
Demand	7	3	5	15 = 15

Total Supply = $5 + 7 + 3 = 15$ = Total Demand = $7 + 3 + 5 = 15$. **Balanced.**

2. Unbalanced Transportation Problem

A transportation problem is **unbalanced** when total supply $\neq$ total demand. Two cases arise:

- **Excess Supply** ($\sum a_i > \sum b_j$): Add a **dummy destination** D_{n+1} with demand $= \sum a_i - \sum b_j$ and zero transportation costs $c_{i,n+1} = 0$. Allocations to the dummy represent surplus (unsent) goods.
- **Excess Demand** ($\sum b_j > \sum a_i$): Add a **dummy source** S_{m+1} with supply $= \sum b_j - \sum a_i$ and zero transportation costs $c_{m+1,j} = 0$. Allocations from the dummy represent unmet (unsatisfied) demand.

After inserting the dummy, the problem becomes balanced and is solved by the standard algorithm.

Example A — Excess Supply ($\sum a_i > \sum b_j$):

	D_1	D_2	D_3	Supply
S_1	5	3	6	20
S_2	4	7	2	30
Demand	15	20	10	

Total Supply = 50, Total Demand = 45, Excess = 5.

Add dummy destination D_4 with demand = 5 and costs = 0:

	D_1	D_2	D_3	D_4 (Dummy)	Supply
S_1	5	3	6	0	20
S_2	4	7	2	0	30
Demand	15	20	10	5	50 = 50

Problem is now balanced and solvable. Units allocated to D_4 remain unsent at the origin.

Example B — Excess Demand ($\sum b_j > \sum a_i$):

	D_1	D_2	D_3	Supply
S_1	2	8	5	10
S_2	4	3	9	15
Demand	10	12	8	

Total Supply = 25, Total Demand = 30, Shortage = 5.

Add dummy source S_3 with supply = 5 and costs = 0:

	D_1	D_2	D_3	Supply
S_1	2	8	5	10
S_2	4	3	9	15
S_3 (Dummy)	0	0	0	5
Demand	10	12	8	30 = 30

Problem is now balanced. Units allocated from S_3 represent demand that cannot be fulfilled.

Comparison Summary:

Type	Condition	Remedy
Balanced	$\sum a_i = \sum b_j$	Solve directly using standard methods (NWCM, VAM, LCM + MODI).
Unbalanced (excess supply)	$\sum a_i > \sum b_j$	Add a dummy destination D_{n+1} with demand = $\sum a_i - \sum b_j$ and zero costs.
Unbalanced (excess demand)	$\sum b_j > \sum a_i$	Add a dummy source S_{m+1} with supply = $\sum b_j - \sum a_i$ and zero costs.